

15330 105....

Decentralized Information Management System

Introduction to the Problem

- Traditional web applications rely on centralized servers for authentication and data storage.
- Centralized systems are vulnerable to data breaches, single points of failure, and unauthorized access.



Solution

- A decentralized information management system that integrates blockchain, MetaMask authentication, and biometric login.
- Key Features:
 - Multi-Authentication: MetaMask, biometric login, and Blockchain.
- By using blockchain, the system ensures security, transparency, and immutability.



Current Gaps in Existing Systems and Our Approach

EXISTING SYSTEM

- CENTRALIZED SYSTEMS → PRONE TO HACKING AND DOWNTIME.
- WEAK USER AUTHENTICATION IN TRADITIONAL APPS.

OUR APPROACH

- ADD BIOMETRIC LOGIN FOR STRONG IDENTITY VERIFICATION.
- USE BLOCKCHAIN-BASED AUTHENTICATION (METAMASK).

Technologies

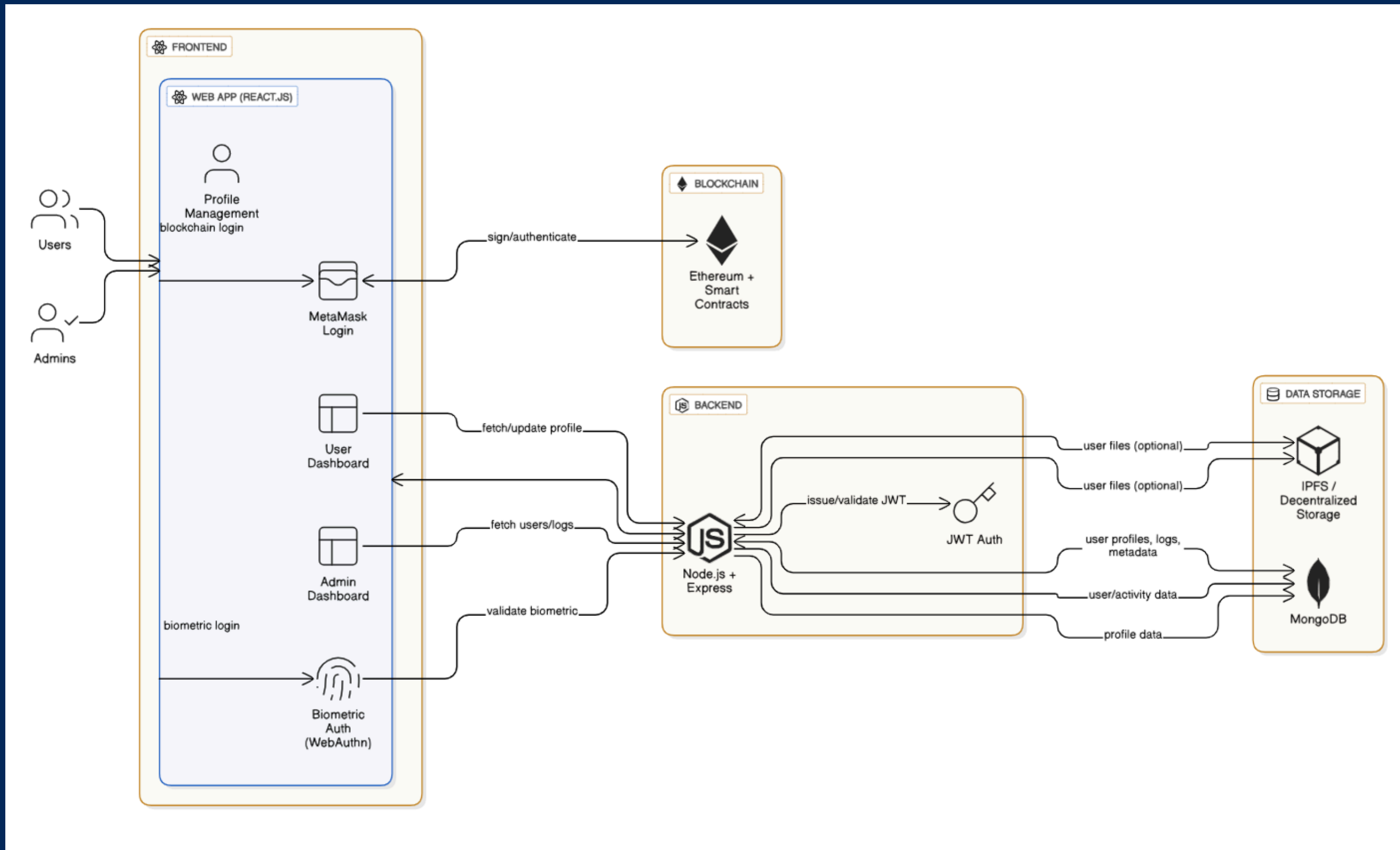
- REACT.JS
- TAILWIND CSS
- NODE.JS + EXPRESS.JS
- MONGODB
- ETHER.JS
- METAMASK
- WEB3.JS
- SMART CONTRACTS (SOLIDITY)
- JWT



Constraints and Limitations

- **Blockchain Dependency:** Requires MetaMask and internet connectivity.
- **Scalability:** High transaction volumes may introduce delays or costs
- **Biometric Hardware:** Users require compatible devices for biometric login.
- **User Adoption:** Some users may find blockchain/MetaMask usage complex.

System Diagram



15330 105....

Thank You !